

Experiment No.	01
Experiment Title	Working with Python packages-Numpy, Scipy, Scikit-Learn, Matplotlib
GitHub Repository	https://github.com/KJayasuriya/ml_laboratory
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1 Objective

- To perform Exploratory Data Analysis (EDA) on the given dataset.
- To find the missing values, outliers, and class distribution.
- To Visualize characteristics of the dataset.

2 Problem Statement

Explore the various functions and methods available in the following Python libraries: Numpy, Pandas, Scipy, Scikit-learn, Matplotlib. Understand the key operations such as array manipulations, data preprocessing, mathematical computing, machine learning workflows, and data visualization [CO1, K2].

3 Dataset Description

Below are the datasets I used for numerical, categorical and Image datasets respectively:

Dataset Name	Pima Indians Diabetes Dataset
Dataset Source	https://archive.ics.uci.edu/ml/datasets/Pima+Indians+Diabetes
Number of Samples	768
Number of Features	9
Number of Classes	2
Missing Values	0

Dataset Name	Loan Prediction
Dataset Source	https://www.kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset
Number of Samples	614
Number of Features	12
Missing Values	149

Dataset Name	MNIST Dataset
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Dataset Source	https://www.kaggle.com/datasets/oddrational/mnist-in-csv
Number of Samples	60000
Number of Features	785
Number of Classes	10
Missing Values	0

4 Exploratory Data Analysis

Loan Prediction

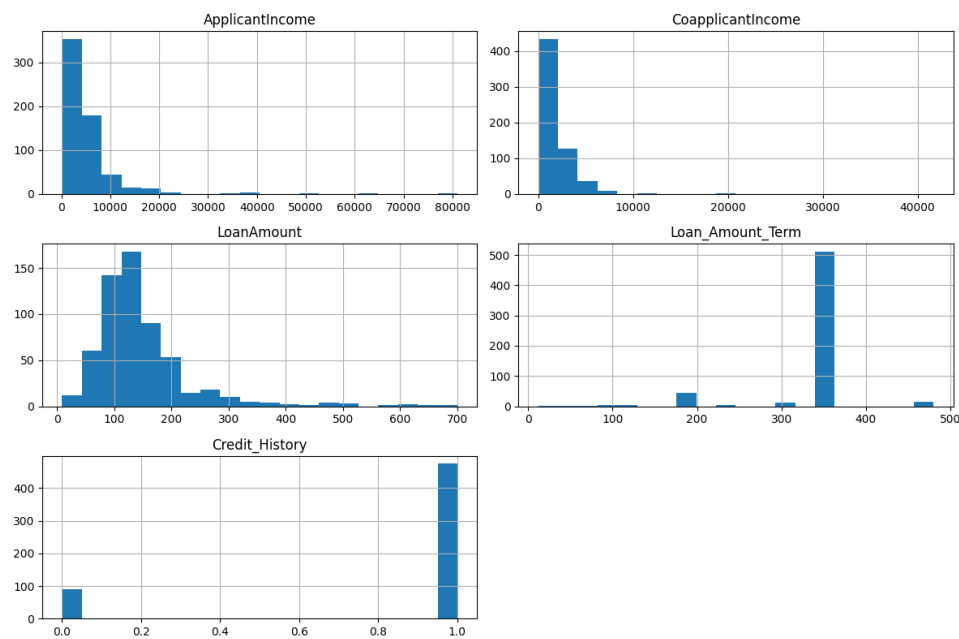


Figure 1: Histogram plots showing statistical summary

Inference: This Histogram plots shows the distribution of each features.

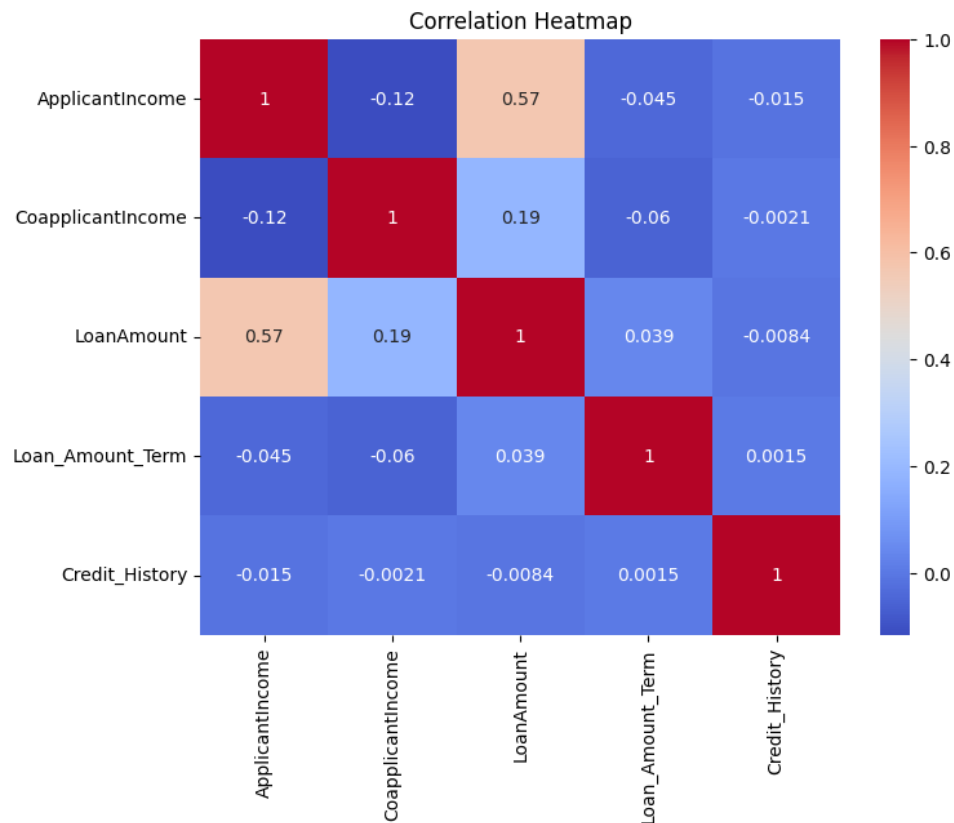


Figure 2: Correlation Matrix

Inference:

The correlation matrix shows the relationships among numerical features. Most features exhibit weak to moderate correlations, indicating that they provide complementary information for prediction.

Diabetes Dataset

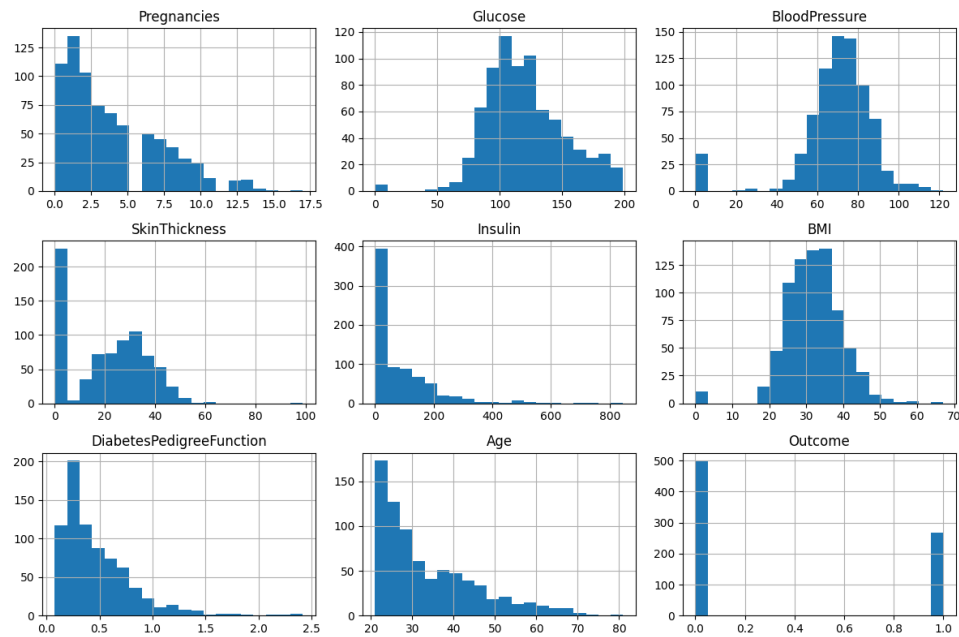


Figure 3: Statistical summary of diabetes dataset

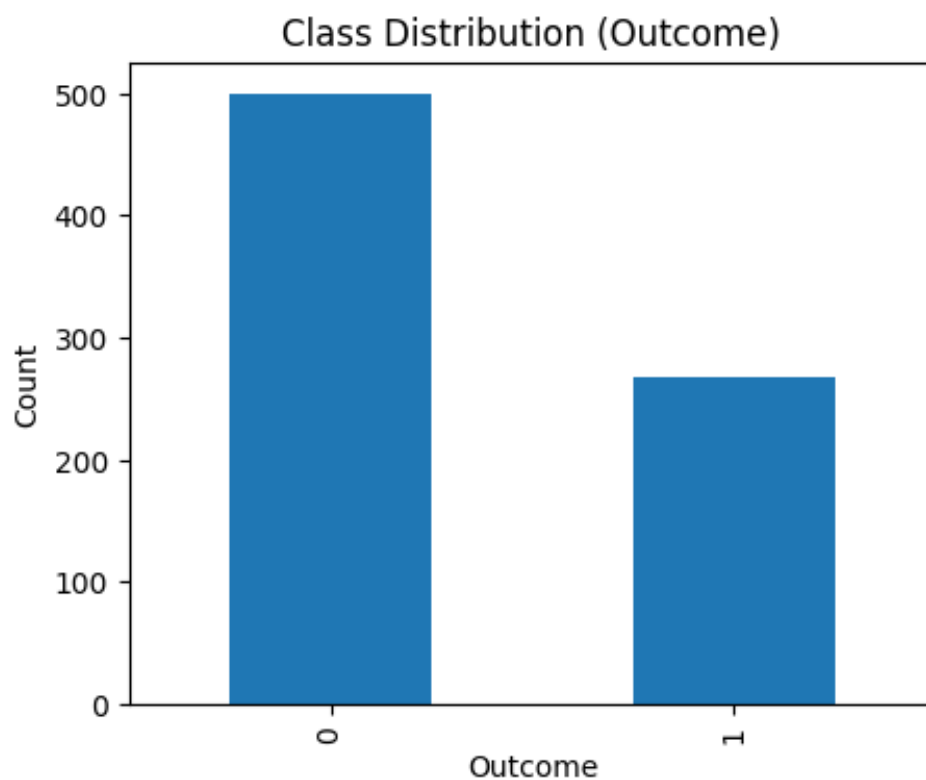


Figure 4: Class Distribution

Inference:

The dataset contains two classes representing diabetic and non-diabetic patients. The class distribution indicates a moderate imbalance, which should be considered during model evaluation.

MNIST Dataset

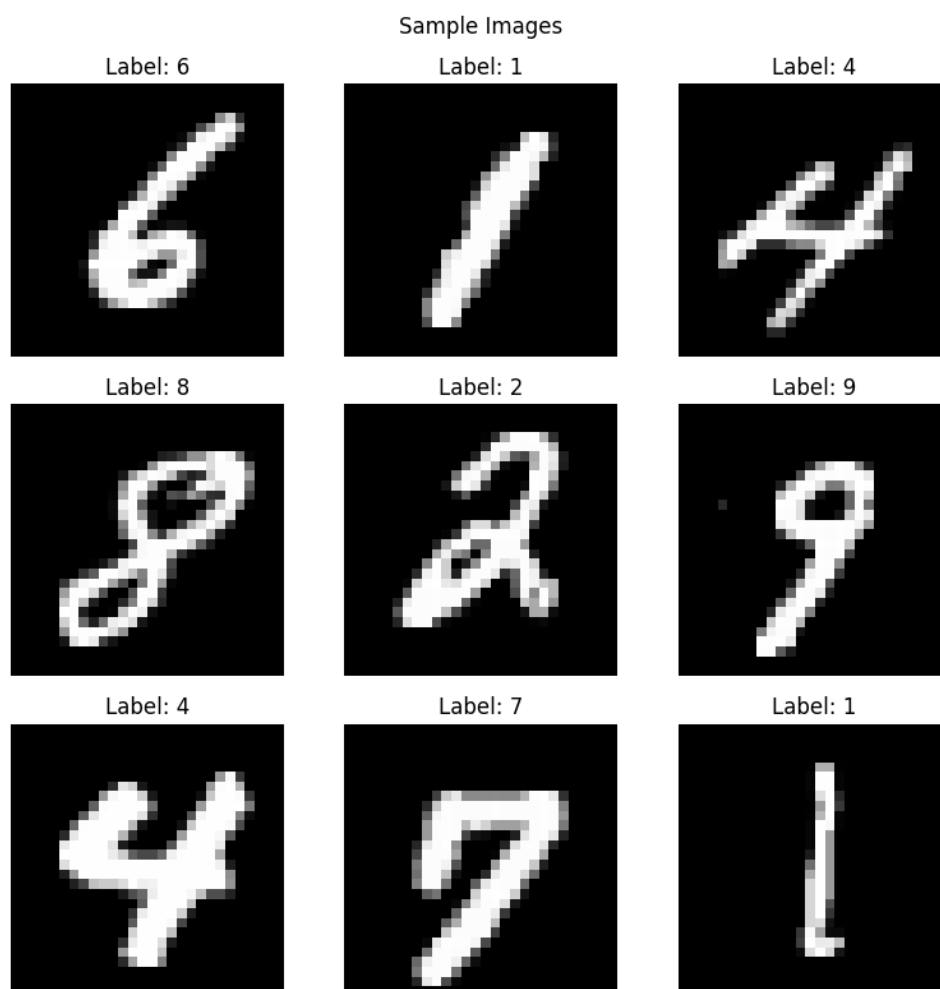


Figure 5: Sample Images from the MNIST Dataset

Inference:

The figure shows randomly selected handwritten digit images from the dataset. Each image is a grayscale image of size 28×28 pixels representing one of the digits from 0 to 9.

5 Experimental Setup

Python Version	3.13.13
Scikit-Learn Version	1.7.5
IDE	Jupyter Notebook
Operating System	Fedora Linux 42
Hardware	Intel core i5 (11th gen) 8GB RAM

6 ML Task Identification

Dataset	Type of ML Task	Feature Selection Technique	Suitable ML Algorithm
Iris Dataset	Classification	ANOVA / SelectKBest	Decision Tree, SVM, KNN
Loan Prediction	Classification	Chi-Square / SelectKBest	Logistic Regression, Random Forest
Predicting Diabetes	Classification	ANOVA / SelectKBest	Logistic Regression, Naive Bayes, KNN
Email Spam Classification	Classification	Chi-Square	Naive Bayes, SVM
MNIST Handwritten Digit Recognition	Classification	PCA (Optional)	KNN, CNN, Random Forest

Table 4: Identification of Machine Learning Tasks

7 Discussion

The experiment provided an overview of the complete machine learning workflow using different types of datasets. Exploratory Data Analysis was performed to understand the characteristics, quality and distribution of the data before model development. Numerical datasets were analyzed using summary statistics, histograms and correlation heatmaps, while categorical data were visualized using bar charts. The MNIST image dataset was explored by displaying sample handwritten digit images. The experiment also helped identify suitable preprocessing techniques, feature selection methods and appropriate machine learning algorithms for different types of datasets.

8 Conclusion

The experiment successfully explored the fundamental Python libraries used in machine learning, including NumPy, Pandas, SciPy, Scikit-learn and Matplotlib. Exploratory Data Analysis was carried out on numerical, categorical and image datasets to understand their structure and characteristics. Different machine learning tasks were identified for each dataset, and suitable preprocessing, feature selection techniques and learning algorithms were studied. This experiment provided a strong foundation for implementing machine learning models in subsequent laboratory experiments.

9 References

1. A.Géron, Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2022.
2. <https://scikit-learn.org/stable/>
3. <https://www.kaggle.com/datasets/somesh24/spambase>